Non-monotonic comoving-frame radiative transfer in CMFGEN

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In the comoving frame, the transfer equation contains a velocity-coupling term linking adjacent frequencies. In monotonic outflows the standard treatment (Hillier & Miller, 1998) absorbs the coupling into a generalised opacity $\hat{\chi} = \chi + \Gamma/\Delta \log \nu$, where $\Gamma = (V/R)(1 + \sigma\mu^2)$ and $\sigma = (R/V)(V/r) - 1$. In interacting supernovae the ejecta contain shocks and compression zones where $\sigma \ll -1$, making $\hat{\chi}$ negative and the formal solution unstable.

Following Knop et al. (2009), we use $|\Gamma|$ in the opacity (keeping it strictly positive) while retaining the signed Γ in the source term $Q = (\Gamma/\Delta \log \nu)/\hat{\chi}$. The upwind frequency is chosen sign-dependently: previous for expanding flow ($\Gamma > 0$), next for compressing flow ($\Gamma < 0$). Since the next-frequency intensity is not yet computed during the sweep, a Gauss–Seidel scheme supplies it from the previous great iteration. The moment equations, unstable for $\sigma < -1$ because the Sobolev resonance has no real root in compression zones, are bypassed; J_ν is computed directly from the formal solution. Each epoch is treated as a static snapshot with the temperature and electron density fixed to the hydrodynamic values. The iteration begins with Lambda iterations from LTE populations with a maximum correction of 10^5 (Hillier & Dessart, 2012), stable with the direct formal solver, and automatically transitions to full linearisation when corrections fall below 50%.

References

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